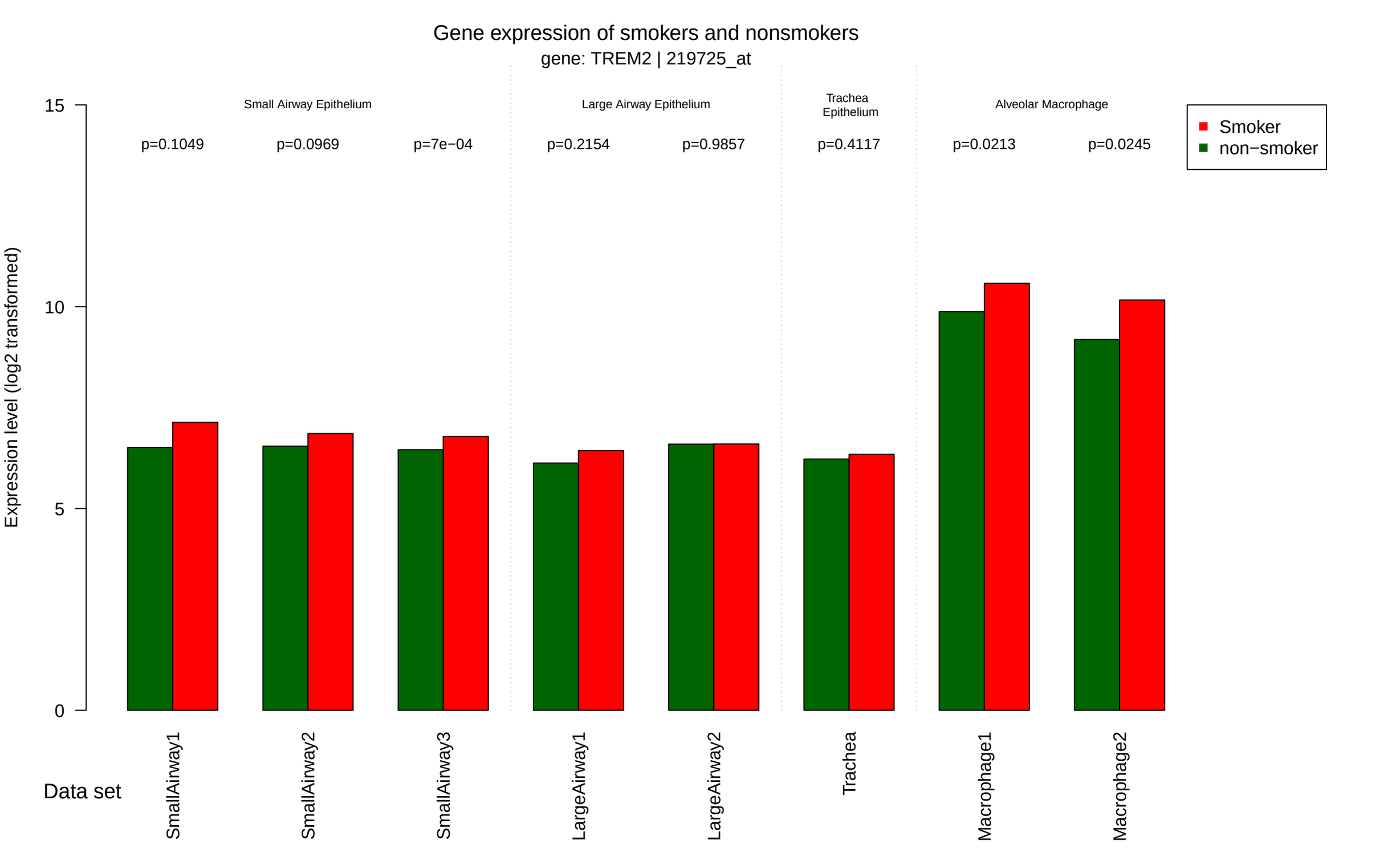
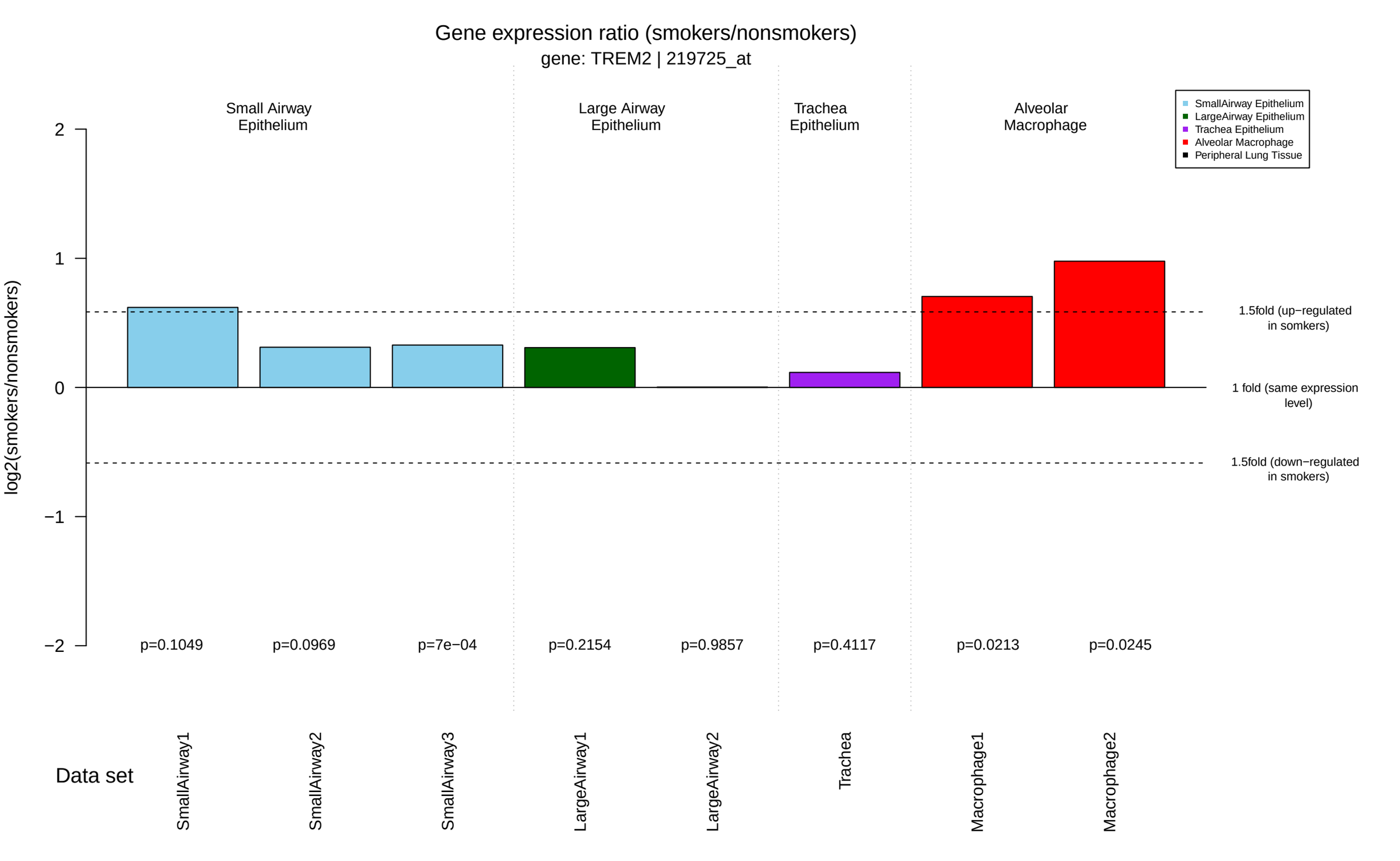
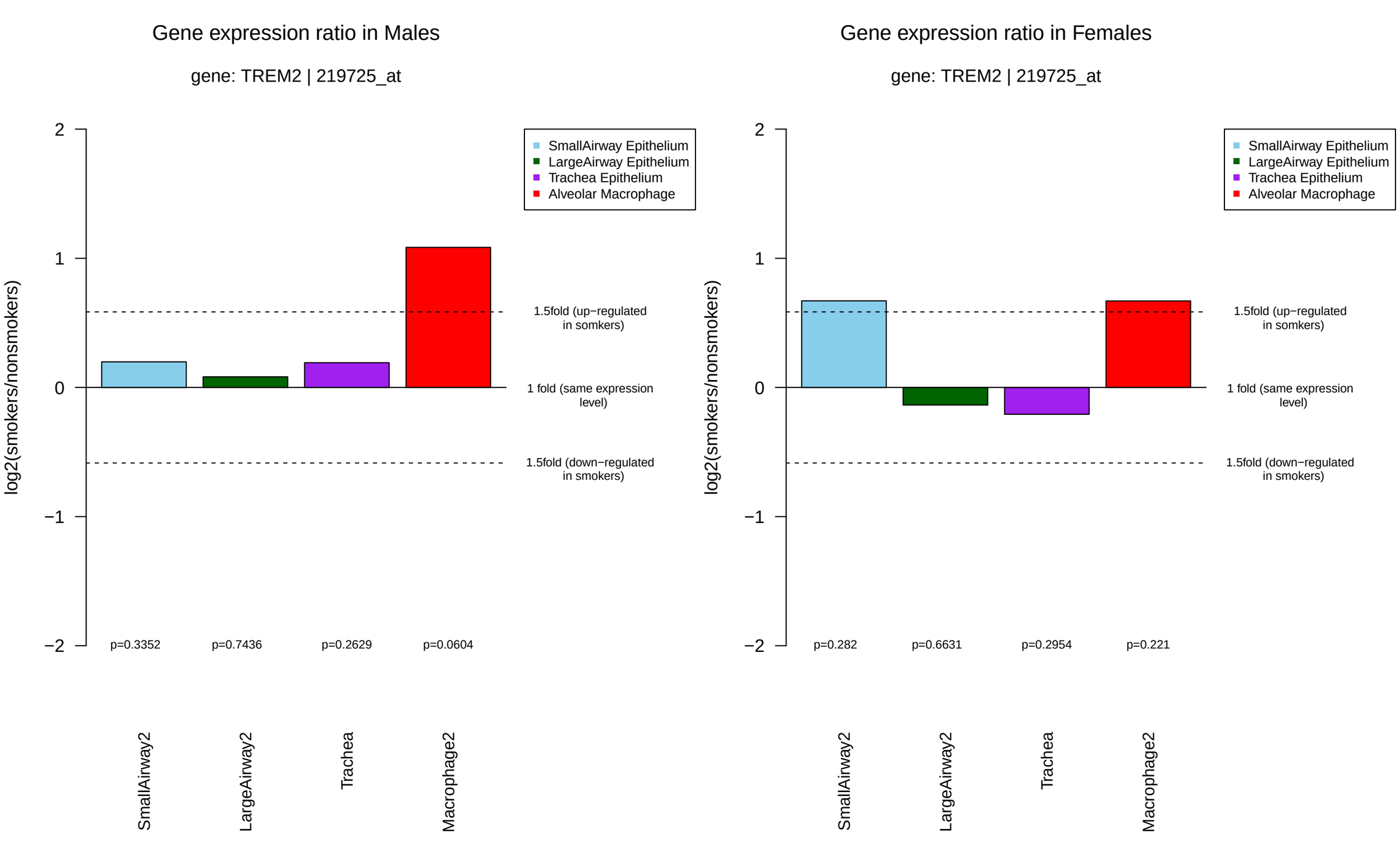
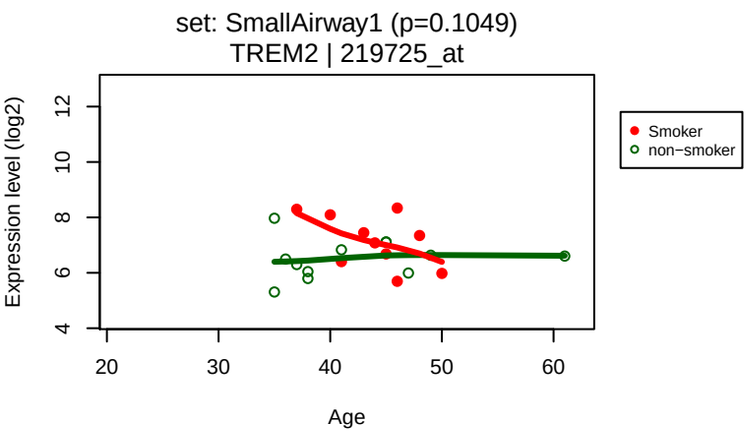
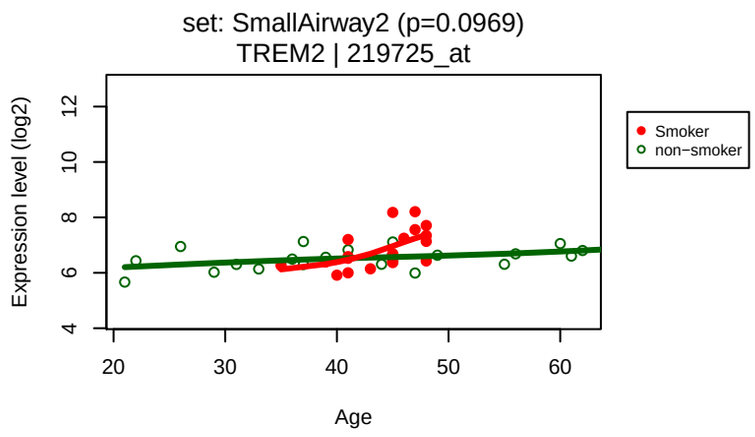
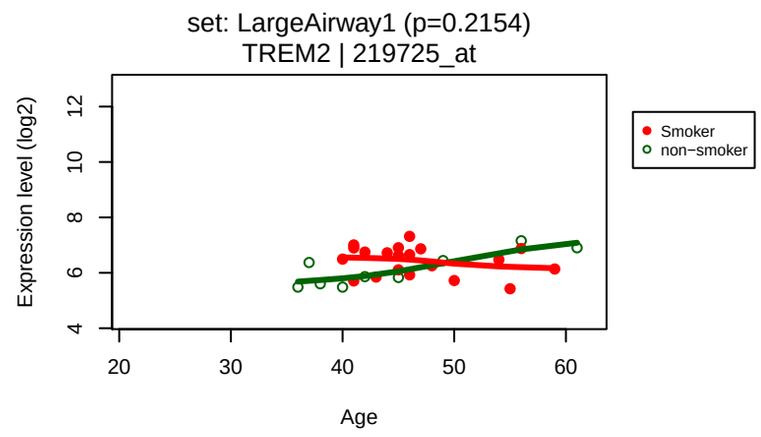

Gene expression of smokers and nonsmokers
gene: TREM2 | 219725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1049
p=0.0969
p=7e−04
p=0.2154
p=0.9857
p=0.4117
p=0.0213
p=0.0245
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

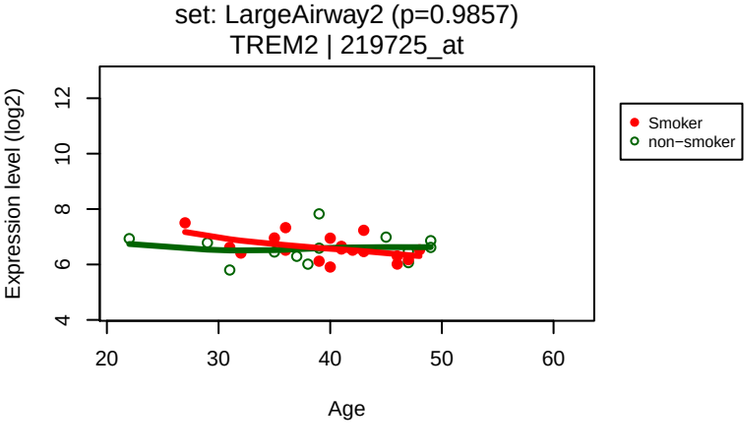

Gene expression ratio (smokers/nonsmokers)
gene: TREM2 | 219725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1049
p=0.0969
p=7e−04
p=0.2154
p=0.9857
p=0.4117
p=0.0213
p=0.0245
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

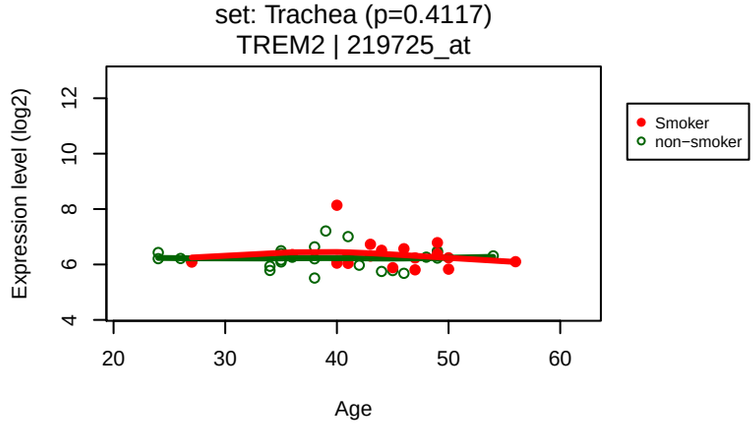

Gene expression ratio in Males
gene: TREM2 | 219725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3352
p=0.7436
p=0.2629
p=0.0604
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TREM2 | 219725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.282
p=0.6631
p=0.2954
p=0.221
SmallAirway2
LargeAirway2
Trachea
Macrophage2

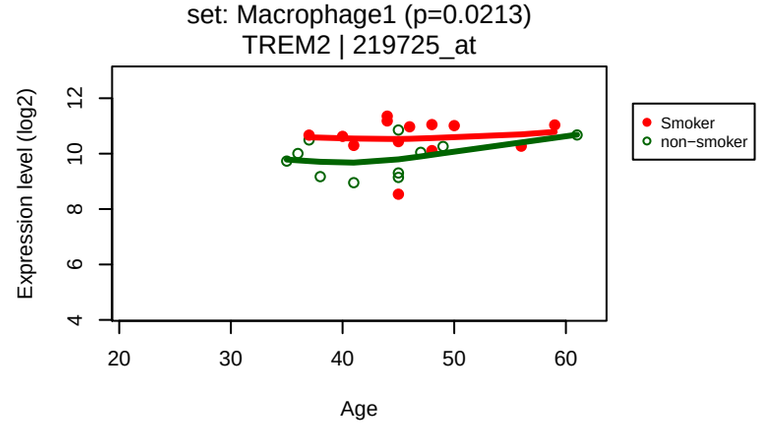

set: SmallAirway1 (p=0.1049)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker

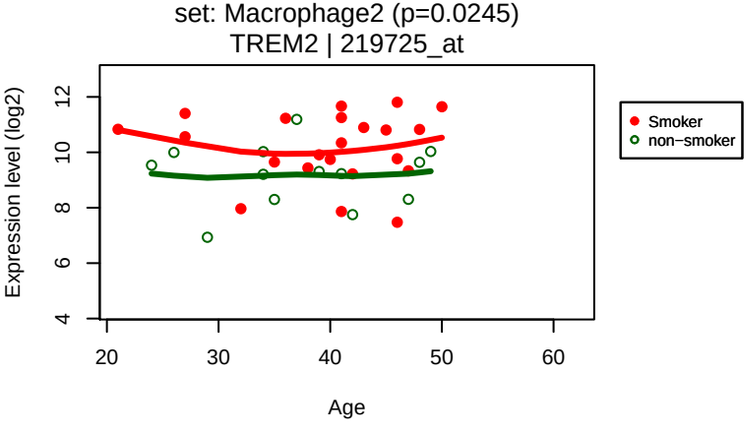

set: SmallAirway2 (p=0.0969)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker

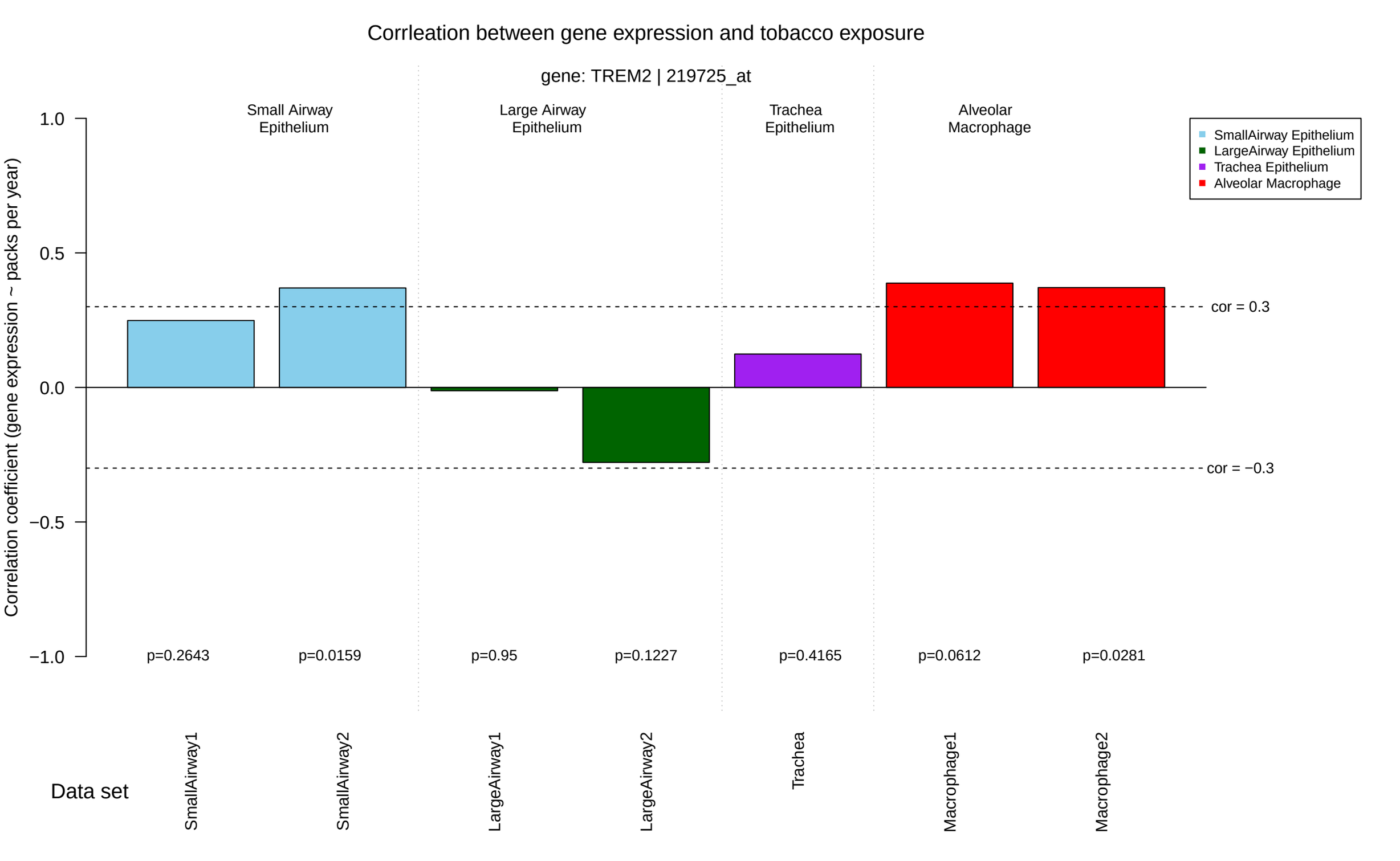

set: LargeAirway1 (p=0.2154)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9857)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4117)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0213)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0245)
TREM2 | 219725_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TREM2 | 219725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2643
p=0.0159
p=0.95
p=0.1227
p=0.4165
p=0.0612
p=0.0281
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2